



Amrita School of Computing

Amritapuri Campus

Department of Computer Science and Engineering

B.Tech Computer Science and Engineering

23CSE498 – Project Phase II

Design and Solution Submission

Project Title Semi-Supervised Learning for Cross-Dataset Generalization in Cervical Cytology Image Classification

Project Guide Ms. Vinitha Panicker

Team Members

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Academic Year: 2026–2027

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1 Sustainable Development Goal (SDG) Mapping

Primary SDG: SDG 3 – Good Health and Well-being

Secondary SDG(s): SDG 9 – Industry, Innovation and Infrastructure

Justification:

This project contributes primarily to SDG 3 by investigating machine learning methods for cervical cytology image classification, with the aim of improving the robustness of computer-aided screening systems. Cervical cancer screening relies on accurate identification of abnormal cellular patterns, while manual examination can be time-consuming and dependent on expert availability. The proposed system investigates Semi-Supervised Learning (Semi-SL), which can utilize large quantities of unlabeled cytology images together with a limited amount of labeled data. The project further considers cross-dataset generalization, where models trained on one dataset are evaluated on independently collected datasets. This addresses the domain shifts that may arise from differences in imaging devices, staining procedures and patient populations. The work also contributes to SDG 9 through the investigation of robust artificial intelligence methods for medical image analysis and the development of a reproducible computational framework for evaluating such methods.

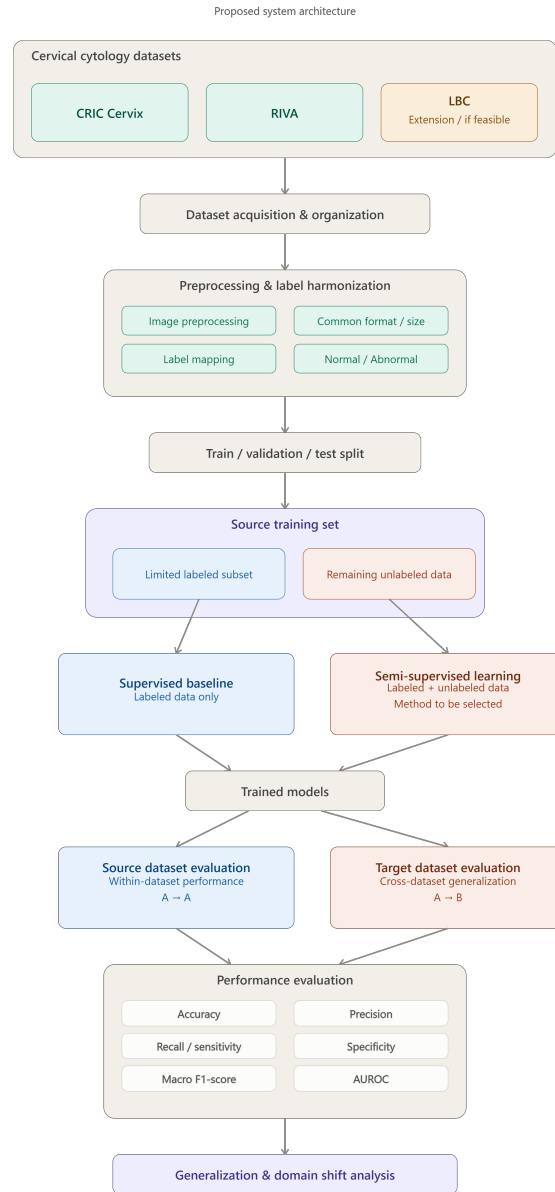
SDG	Goal
1	No Poverty
2	Zero Hunger
3	Good Health and Well-being
4	Quality Education
5	Gender Equality
6	Clean Water and Sanitation
7	Affordable and Clean Energy
8	Decent Work and Economic Growth
9	Industry, Innovation and Infrastructure
10	Reduced Inequalities
11	Sustainable Cities and Communities
12	Responsible Consumption and Production
13	Climate Action
14	Life Below Water
15	Life on Land
16	Peace, Justice and Strong Institutions
17	Partnerships for the Goals

2 Project Milestones and Timeline

Semester	Major Activity / Milestone	Expected Deliverable	Expected Completion	Status
S6	Project domain exploration	Problem domain identified	15 Feb 2026	Completed
S6	Literature review	Initial literature survey	15 Mar 2026	Completed
S6	Baseline implementation	Reference implementation completed	30 Apr 2026	Completed
S7	Literature survey	Comprehensive literature survey	15 Jul 2026	Completed
S7	Research gap identification	Research gap statement	20 Jul 2026	Completed
S7	Problem statement	Finalized objectives	25 Jul 2026	Completed
S7	System design	Architecture, workflow and methodology	9 Aug 2026	Current
S7	Prototype implementation	Functional prototype	15 Oct 2026	Planned
S7	Module integration	Integrated experimental system	15 Nov 2026	Planned
S7	Testing and validation	Experimental results	15 Dec 2026	Planned
S8	System enhancement	Improved system	31 Jan 2027	Planned
S8	Performance evaluation	Performance analysis	28 Feb 2027	Planned
S8	Research publication	Paper submission	20 Mar 2027	Planned
S8	Dissertation writing	Final report	10 Apr 2027	Planned
S8	Final demonstration	Presentation and viva	20 Apr 2027	Planned

Table 1: Overall Project Timeline

3 System Architecture



Architecture Description

The proposed system is designed as a modular machine learning pipeline for investigating the effect of Semi-Supervised Learning on cross-dataset generalization in cervical cytology image classification.

The system begins with the acquisition of multiple publicly available cervical cytology datasets. The datasets undergo preprocessing and label harmonization so that the data can be evaluated using a common classification setting. Each dataset is divided into training, validation and test subsets. Within the source-domain training set, only a

limited fraction of samples is treated as labeled data, while the remaining training samples are treated as unlabeled data for the Semi-SL setting.

Two training branches are considered. The first branch provides a supervised baseline trained only using the available labeled samples. The second branch uses the same labeled samples together with the available unlabeled samples through a suitable Semi-SL framework.

The trained models are evaluated both on the source dataset and on independent target datasets. This allows the study to distinguish within-dataset performance from cross-dataset generalization performance. The final evaluation module compares the supervised and Semi-SL approaches using appropriate classification metrics and analyzes the effect of dataset domain shift.

4 Module Description

4.1 Dataset Acquisition Module

Purpose: To obtain and organize the publicly available cervical cytology datasets used for the study.

Inputs:

- Public cervical cytology datasets
- Dataset metadata and class labels

Outputs:

- Organized image datasets
- Corresponding metadata and labels

Functionality: The module collects the selected datasets and organizes their images, labels and associated metadata into a consistent directory and data representation suitable for subsequent processing.

4.2 Data Preprocessing and Label Harmonization Module

Purpose: To prepare images from different datasets for common model training and evaluation.

Inputs:

- Raw cervical cytology images
- Original dataset-specific labels

Outputs:

- Preprocessed images
- Harmonized class labels

Functionality: The module applies common preprocessing operations required by the selected classification models and maps dataset-specific labels into a common classification taxonomy. This step is necessary because the datasets may use different class definitions and data formats.

4.3 Dataset Splitting and Limited-Label Setup Module

Purpose: To establish controlled training, validation and testing partitions and simulate limited availability of labeled data.

Inputs:

- Harmonized dataset

Outputs:

- Training set, validation set, test set
- Labeled training subset
- Unlabeled training subset

Functionality: Each dataset is first divided into training, validation and test subsets. The source training set is then used to construct limited label settings. In the core experimental setting, a small fraction such as 5% of the training data is treated as labeled, while the remaining training samples are treated as unlabeled for Semi-SL training. Additional label fractions may be considered during the implementation and evaluation stages.

4.4 Supervised Baseline Module

Purpose: To establish a baseline for comparison with Semi-SL.

Inputs:

- Labeled source-domain training data

Outputs:

- Trained supervised classification model

Functionality: The supervised baseline is trained using only the available labeled training samples. The same model evaluation protocol is subsequently used to measure its performance on source and target datasets.

4.5 Semi-Supervised Learning Module

Purpose: To exploit both labeled and unlabeled source-domain training data.

Inputs:

- Labeled training data
- Unlabeled training data

Outputs:

- Trained Semi-SL classification model

Functionality: The module applies a suitable Semi-SL framework to learn from the limited labeled data and the remaining unlabeled data. Candidate Semi-SL approaches will be studied during implementation and the final method will be selected based on suitability for the datasets, computational feasibility and experimental performance.

4.6 Evaluation and Generalization Analysis Module

Purpose: To evaluate classification performance and investigate cross-dataset generalization.

Inputs:

- Trained supervised model
- Trained Semi-SL model
- Source test data
- Target test data

Outputs:

- Classification metrics
- Comparison tables
- Performance plots
- Cross-dataset generalization analysis

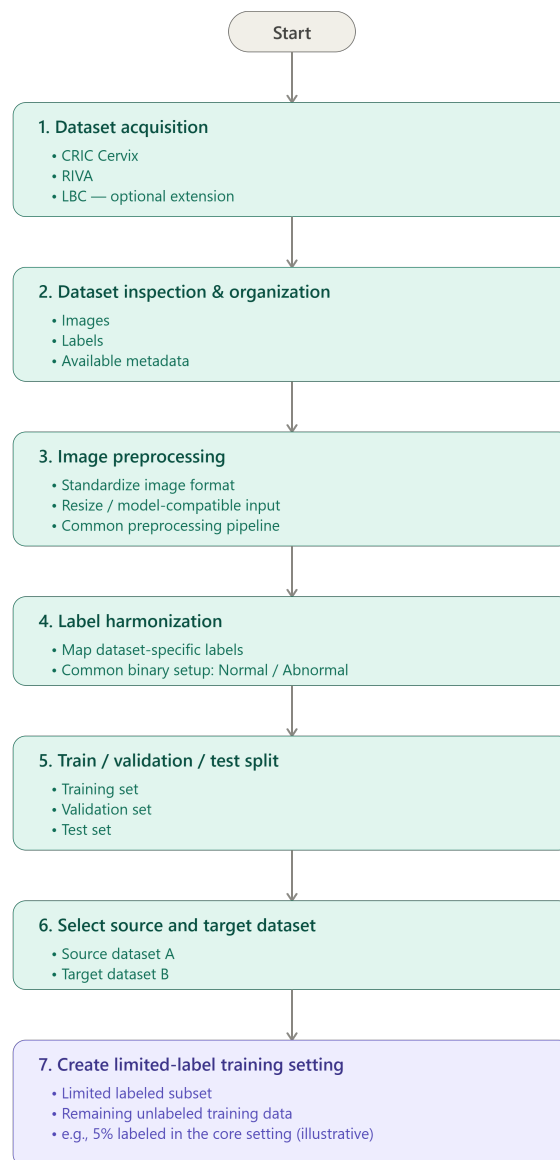
Functionality: The module evaluates both approaches under consistent testing conditions. Source-domain evaluation measures within-dataset performance, while target-domain evaluation measures generalization to an independently collected dataset. The results are compared to determine whether the use of unlabeled data provides an advantage under dataset domain shift.

5 System Workflow

The proposed workflow consists of fifteen stages, presented below in three parts for clarity: data preparation, the parallel training branches with validation, and final evaluation with generalization analysis.

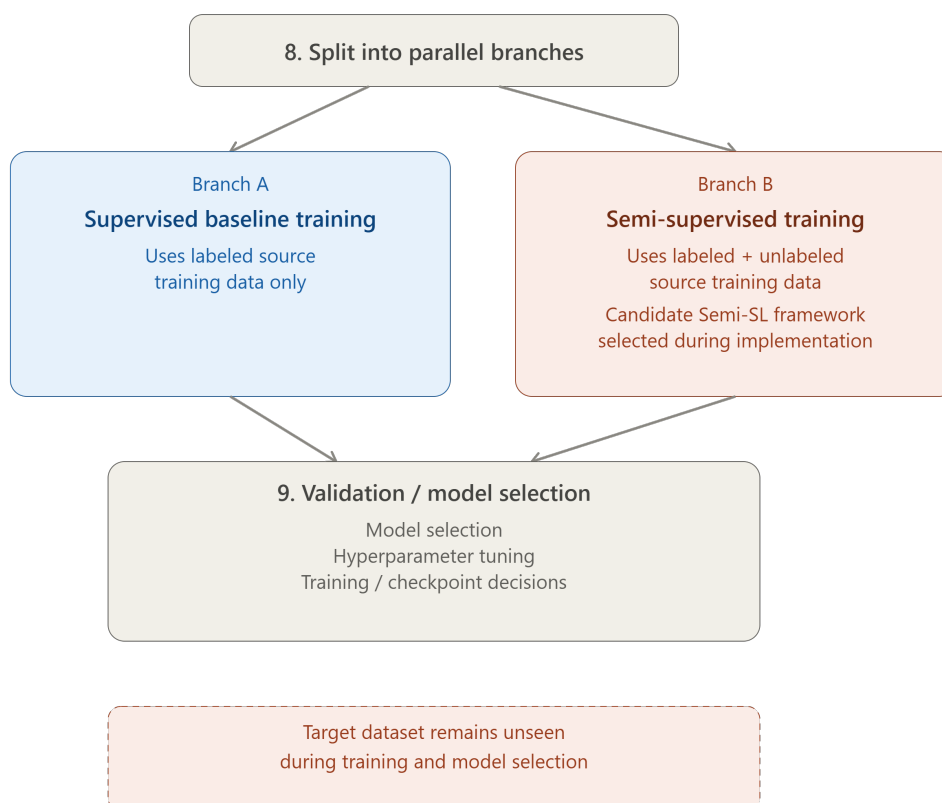
Part 1 – Data Preparation (Steps 1–7)

The first part covers everything from acquiring the raw cervical cytology datasets through to constructing the limited-label training setup: dataset acquisition, inspection and organization, image preprocessing, label harmonization, the train/validation/test split, selection of the source and target datasets, and creation of the limited-label training setting.



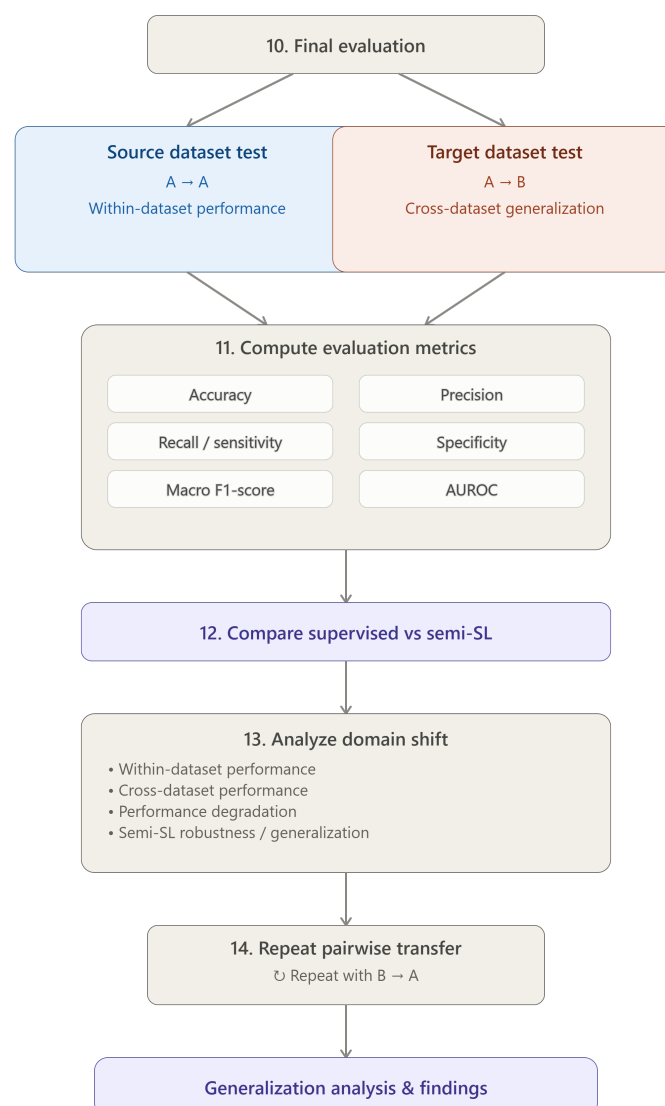
Part 2 – Parallel Training Branches and Validation (Steps 8–9)

From the limited-label training setting, the process splits into the two parallel branches being compared – the supervised baseline and the Semi-SL model – both of which then feed into validation and model selection. The target dataset is deliberately kept unseen throughout training and validation.



Part 3 – Evaluation and Generalization Analysis (Steps 10–14)

After validation, the trained models are tested on both the source and target datasets, the evaluation metrics are computed, the two training strategies are compared, the domain-shift analysis is carried out, and the pairwise source-target evaluation is repeated for the remaining dataset combinations.



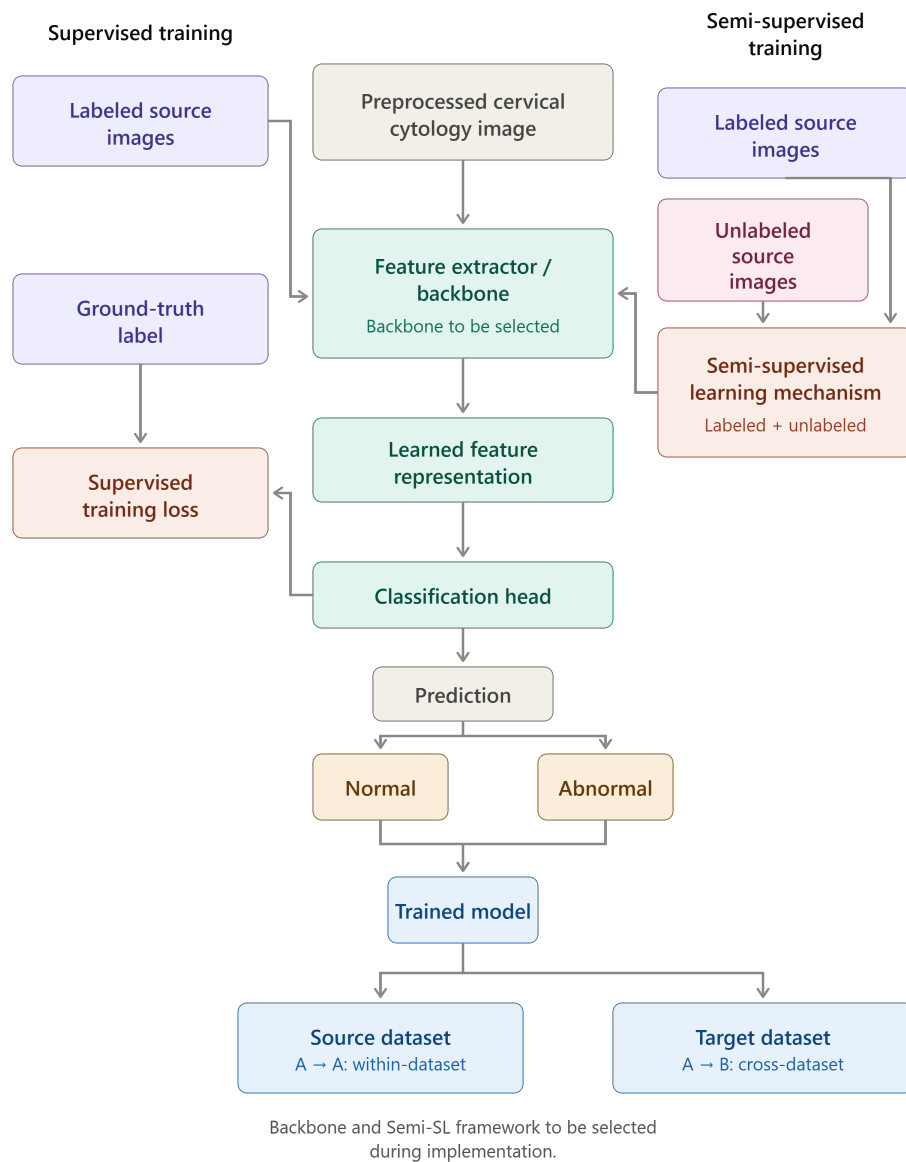
6 Domain-Specific Design

6.1 Dataset Description

The project will use publicly available cervical cytology datasets for investigating limited-label learning and cross-dataset generalization.

Dataset	Role	Description	Planned Use
CRIC Cervix	Primary	Conventional Pap smear cervical cell images with annotated cells	Source/Target
RIVA	Primary	Cervical cytology images collected across multiple diagnostic categories	Source/Target
LBC	Secondary / Extension	Liquid-Based Cytology dataset	Extension if feasible

6.2 Machine Learning Model Architecture



The proposed classification architecture consists of an image feature extraction stage followed by a classification stage. A suitable pre-trained or lightweight deep learning backbone will be considered for feature extraction, followed by a classification head appropriate for the harmonized classification task.

The final backbone will be selected during implementation after considering dataset characteristics, computational requirements and experimental suitability.

For the Semi-SL setting, the classification architecture will be integrated with a suitable Semi-SL training framework capable of utilizing both labeled and unlabeled training samples.

The system is therefore designed independently of a single fixed backbone or Semi-SL algorithm at this stage, allowing candidate methods to be evaluated during the prototype implementation phase.

6.3 Experimental Setting

The study will use a controlled limited-label setting. For a source dataset, the training, validation and test partitions are used as follows:

Partition	Role in the Experimental Setting
Training data – Limited labeled subset	Used to train both the supervised baseline and the Semi-SL model.
Training data – Remaining unlabeled subset	Used only by the Semi-SL model, in addition to the labeled subset.
Validation data	Used for model selection and hyperparameter tuning.
Test data	Used only for the final source- and target-domain evaluation; never used during training or tuning.

The same source dataset is used to establish the supervised and Semi-SL training conditions, while independent datasets are used to evaluate cross-dataset generalization.

The core study will focus on pairwise source-to-target transfer settings. Additional label fractions and datasets may be considered depending on implementation feasibility and available time.

7 Algorithms / Methodology

7.1 Overall Methodology

The proposed methodology investigates whether Semi-SL can improve classification performance and robustness when the model is trained with limited labeled data and evaluated under dataset domain shift.

The methodology consists of the following stages:

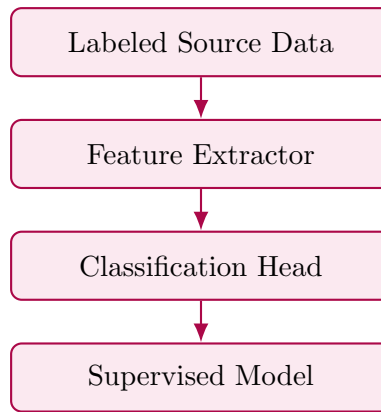
Step 1: Collect and organize the selected cervical cytology datasets.

- Step 2:** Preprocess the images using a common preprocessing pipeline.
- Step 3:** Harmonize the class labels across the selected datasets.
- Step 4:** Divide each dataset into training, validation and test subsets.
- Step 5:** Select a source dataset and a target dataset for a pairwise transfer experiment.
- Step 6:** Within the source training set, select the required fraction of labeled samples and treat the remaining samples as unlabeled.
- Step 7:** Train a supervised baseline using only the labeled source training samples.
- Step 8:** Train a Semi-SL model using both the labeled and unlabeled source training samples.
- Step 9:** Use the validation set for model selection and hyperparameter adjustment without using the target test set.
- Step 10:** Evaluate the supervised and Semi-SL models on the source test set.
- Step 11:** Evaluate the same trained models on the independent target dataset.
- Step 12:** Calculate the selected evaluation metrics.
- Step 13:** Compare supervised and Semi-SL performance under both within-dataset and cross-dataset settings.
- Step 14:** Analyze whether Semi-SL reduces the performance degradation caused by dataset domain shift.

7.2 Supervised Baseline

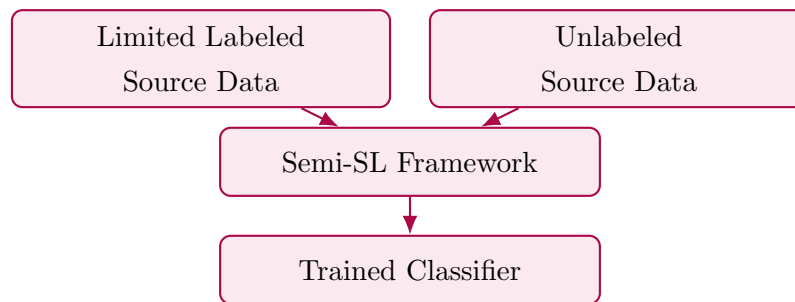
The supervised baseline will use only the labeled subset of the source-domain training data. The baseline provides a controlled reference against which the Semi-SL approach can be evaluated.

Conceptually:



7.3 Semi-Supervised Learning

The Semi-SL setting uses both the labeled and unlabeled portions of the source training data together:



The specific Semi-SL framework will be selected during the implementation stage after evaluating candidate approaches suitable for the project.

7.4 Pairwise Cross-Dataset Evaluation

For two datasets A and B:

Experiment 1: Train on A $\rightarrow$ Test on A

Experiment 2: Train on A $\rightarrow$ Test on B

The same protocol is applied to both the supervised and Semi-SL training conditions.

For multiple datasets, the source-target experiment can be repeated for the relevant unique dataset pairs, for example:

$A \rightarrow B$, $B \rightarrow A$

and, where a third dataset is included:

$A \rightarrow C$, $C \rightarrow A$, $B \rightarrow C$, $C \rightarrow B$

7.5 Evaluation Metrics

The study will use classification metrics appropriate for medical image classification and class imbalance. The primary metrics considered are:

- Accuracy
- Precision
- Recall / Sensitivity
- Specificity
- Macro F1-score
- AUROC

Recall / sensitivity is particularly important because false-negative predictions may correspond to abnormal cells being classified as normal. Macro F1-score is useful for ensuring that performance across classes is represented rather than allowing a dominant class to determine the overall score.

The final set of primary and secondary metrics will be fixed during the experimental phase based on the final class configuration and evaluation requirements.

7.6 Pseudocode

Algorithm: Cross-Dataset Semi-Supervised Evaluation

Input: Multiple cervical cytology datasets $D_1, D_2, \dots, D_n$; labeled fraction r ; candidate Semi-SL method

Output: Within-dataset and cross-dataset performance results

1. Load selected datasets.
2. Preprocess images.
3. Harmonize class labels.
4. For each source dataset S :
 - (a) Split S into training, validation and test sets.
 - (b) Select labeled fraction r from S_{train} .
 - (c) Treat remaining S_{train} samples as unlabeled.
 - (d) Train supervised model using labeled S_{train} .
 - (e) Train Semi-SL model using labeled and unlabeled S_{train} .
 - (f) Evaluate both models on S_{test} .
 - (g) For each target dataset $T \neq S$:
 - i. Evaluate both trained models on T_{test} .
 - ii. Compute evaluation metrics.

- iii. Record source-domain and target-domain performance.
5. Compare supervised and Semi-SL results.
6. Analyze cross-dataset performance degradation.
7. Report generalization findings.

8 Technology Stack

Category	Technology / Tool
Programming Language	Python
Deep Learning Framework	PyTorch
Image Processing	OpenCV, Pillow
Numerical Computing	NumPy
Data Handling	Pandas
Machine Learning / Evaluation	scikit-learn
Development Environment	Jupyter Notebook / VS Code
Version Control	Git
Repository	GitHub
Dataset Storage	Local / Cloud storage as required
Experiment Environment	Local GPU or cloud GPU environment, if required

The technology stack is selected to support image preprocessing, deep learning model development, Semi-SL experimentation, quantitative evaluation and reproducible experiment management.

9 Hardware and Software Requirements

9.1 Hardware Requirements

Component	Requirement
Processor	Modern multi-core CPU
RAM	Minimum 16 GB recommended
Storage	Sufficient local storage for multiple image datasets and model checkpoints
GPU	CUDA-compatible NVIDIA GPU recommended for deep learning training
Internet Connectivity	Required for dataset acquisition, package installation and repository access
Other Hardware	Not specifically required

9.2 Software Requirements

Software	Version / Details
Operating System	Windows / Linux
Programming Language	Python 3.x
Deep Learning Framework	PyTorch
Image Processing	OpenCV, Pillow
Numerical Computing	NumPy
Data Processing	Pandas
Evaluation	scikit-learn
IDE / Editor	VS Code / Jupyter
Version Control	Git
Repository	GitHub
Other Dependencies	CUDA and compatible GPU drivers where GPU acceleration is used

10 Design Considerations (Non-Functional)

10.1 Performance

The system should support efficient training and evaluation of multiple source-target dataset combinations. Computationally appropriate model architectures will be considered to make repeated Semi-SL experiments feasible within the project resources.

10.2 Reliability

The supervised and Semi-SL approaches will be evaluated under consistent preprocessing, dataset partitioning and testing conditions. The target test data will remain separate from the training and validation process to avoid data leakage.

10.3 Reproducibility

Random seeds, dataset partitions, label fractions, model configurations and evaluation results should be recorded wherever possible. The implementation will be organized into separate preprocessing, training and evaluation components to facilitate repeated experiments.

10.4 Maintainability

The system will use modular components for dataset loading, preprocessing, training and evaluation. This will allow candidate backbones and Semi-SL methods to be changed without redesigning the entire pipeline.

10.5 Scalability

The proposed architecture can be extended to additional cervical cytology datasets and different labeled-data fractions without fundamentally changing the evaluation pipeline.

10.6 Privacy

The project uses publicly available datasets and does not involve direct collection of new patient data. Any available metadata will be handled only as required for the research task.

10.7 Ethical Considerations

The proposed system is intended as a research and evaluation framework and is not a clinically validated diagnostic system. Experimental results should therefore not be interpreted as medical diagnoses. The study aims to investigate computational robustness under limited-label and cross-dataset settings rather than replace clinical expertise.

11 Expected Outcome

The expected outcomes of the project are:

1. A standardized preprocessing and label-harmonization pipeline for the selected cervical cytology datasets.
2. A modular deep learning framework for training supervised and Semi-SL classification models under limited labeled-data settings.
3. A controlled evaluation protocol for within-dataset and cross-dataset testing.
4. A supervised baseline that uses only the available labeled training samples.
5. A Semi-SL model that additionally utilizes the available unlabeled training data.
6. Quantitative comparisons between supervised and Semi-SL learning under source-domain and cross-dataset evaluation.
7. An analysis of the effect of dataset domain shift on cervical cytology classification performance.
8. An assessment of whether Semi-SL provides measurable benefits for cross-dataset generalization when labeled data are limited.
9. A reproducible codebase containing the dataset processing, training and evaluation components of the proposed framework.

12 References

References

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Faculty Remarks

Guide Remarks

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